

Section: Module 2

Permutation Test

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GOV 2003

September 18th, 2026

Today's Agenda

- Permutation Test
 - Simple example under sharp null
 - More general sharp null
 - ① Constant treatment effect
 - ② Effect increasing
 - Test Inversion
- Conditional Independence Test
- Review Question

Permutation Test: Review

- Finite population framework
 - Only source of randomness is **treatment assignment**
 - We fix the potential outcome $\mathcal{O}_n = \{Y_i(1), Y_i(0)\}_{i=1}^n$
- Assume **Unconfoundedness**
 - $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp T_i$
 - Guaranteed by design (RCT) $\rightarrow$ we can permute the treatment
- Overview of Hypothesis Testing Procedure
 - Assume the null hypothesis
 - Permute treatment assignment under the null and calculate the test statistic
 - If we see that the observed test statistic is significantly different from the ones calculated under the null, then it indicates the null hypothesis is wrong

Permutation Test: Example

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	3	?	?
2	1	5	?	?
3	0	2	?	?
4	0	1	?	?

- Calculate the observed test statistic

$$\begin{aligned} S &= \sum_{i=1}^4 T_i R_i(Y_1, Y_2, Y_3, Y_4) \\ &= 3 \times 1 + 4 \times 1 + 2 \times 0 + 1 \times 0 \\ &= 7 \end{aligned}$$

Permutation Test: Example

- Let's fill the table.
- By using $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$ (consistency)

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	3	3	?
2	1	5	5	?
3	0	2	?	2
4	0	1	?	1

Permutation Test: Example

- Now, let's assume sharp null $H_0 : Y_i(1) = Y_i(0)$
- Under the sharp null, we can fill the missing values.

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	3	3	3
2	1	5	5	5
3	0	2	2	2
4	0	1	1	1

Permutation Test: Example

- Under the sharp null, let's permute treatment assignment
- There are six combinations of treatment assignment
 - $(1,1,0,0)$
 - $(1,0,1,0)$
 - $(1,0,0,1)$
 - $(0,1,1,0)$
 - $(0,1,0,1)$
 - $(0,0,1,1)$

Permutation Test: Example

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$	$R_i(\mathbf{Y}(1))$	$R_i(\mathbf{Y}(0))$
1	1	3	3	3	3	3
2	1	5	5	5	4	4
3	0	2	2	2	2	2
4	0	1	1	1	1	1

- Compute the test statistic under the permuted assignment
 - $(1,1,0,0) \rightarrow 3 + 4 = 7$
 - $(1,0,1,0) \rightarrow 3 + 2 = 5$
 - $(1,0,0,1) \rightarrow 3 + 1 = 4$
 - $(0,1,1,0) \rightarrow 4 + 2 = 6$
 - $(0,1,0,1) \rightarrow 4 + 1 = 5$
 - $(0,0,1,1) \rightarrow 2 + 1 = 3$

More general sharp null (1): Constant treatment effect

- Consider more general sharp null $H_0 : Y_i(1) = Y_i(0) + \tau_0$
 - Let's say, $\tau_0 = 5$
- Let's think about the same toy example

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	3	?	?
2	1	5	?	?
3	0	2	?	?
4	0	1	?	?

More general sharp null (1): Constant treatment effect

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	3	?	?
2	1	5	?	?
3	0	2	?	?
4	0	1	?	?

- In observed data, treatment effect adjusted Wilcoxon's rank sum test statistic is given by

$$\begin{aligned} S &= \sum_{i=1}^4 T_i R_i(Y_1 - T_1 \tau_0, Y_2 - T_2 \tau_0, Y_3 - T_3 \tau_0, Y_4 - T_4 \tau_0) \\ &= 1 \times 1 + 2 \times 1 + 4 \times 0 + 3 \times 0 \\ &= 3 \end{aligned}$$

More general sharp null (2): Constant treatment effect

- Let's fill the table. With consistency, we can impute $Y_i(T_i)$

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	3	3	?
2	1	5	5	?
3	0	2	?	2
4	0	1	?	1

More general sharp null (3): Constant treatment effect

- Then, let's use sharp null $H_0 : Y_i(1) = Y_i(0) + \tau_0$ with $\tau_0 = 5$

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	3	3	-2
2	1	5	5	0
3	0	2	7	2
4	0	1	6	1

More general sharp null (4): Constant treatment effect

- Under the sharp null, let's permute treatment assignment and calculate the test statistic
- Notice that, under the sharp null, the test statistic is

$$\begin{aligned} S &= \sum_{i=1}^4 T_i R_i(Y_1 - T_1\tau_0, Y_2 - T_2\tau_0, Y_3 - T_3\tau_0, Y_4 - T_4\tau_0) \\ &= \sum_{i=1}^4 T_i \underbrace{R_i(Y_1(0), Y_2(0), Y_3(0), Y_4(0))}_{\text{Does not depend on } T_i} \end{aligned}$$

- Since we fill $Y_i(0)$ under the sharp null, we just need to permute the treatment assignment T_i

More general null (5): Effect increasing

- Suppose that you want to test

$$H_0 : Y_i(1) - Y_i(0) \leq 3 \quad \text{for all } i$$

- Unlike a sharp null, we cannot exactly fill the missing values, but we can fill their possible ranges.

ID	T_i	Y_i	$Y_i(1)$	$Y_i(0)$
1	1	4	4	≥ 1
2	1	7	7	≥ 4
3	0	3	≤ 6	3
4	0	2	≤ 5	2

- There are many possible tables that satisfy the null hypothesis.

More general null (6): Effect increasing

- Consider two possible tables that satisfy

$$H_0 : Y_i(1) - Y_i(0) \leq 3.$$

	$\mathbf{Y}(0)$	Treatment effects	Rank sum S
Boundary	(1, 4, 3, 2)	(3, 3, 3, 3)	$1 + 4 = 5$
Another possibility	(4, 5, 3, 2)	(0, 2, 3, 3)	$3 + 4 = 7$

- For treated units,

$$Y_i(0) = Y_i(1) - \{Y_i(1) - Y_i(0)\} \geq Y_i(1) - 3.$$

- Thus, setting the treatment effect to exactly 3 gives the **smallest possible** $Y_i(0)$ **for treated units**.
- Any smaller treatment effect moves treated $Y_i(0)$ upward and cannot decrease their rank sum.

More general null (7): Effect increasing

- Although different values inside the range give different observed test statistics, the reference distribution is the same.
- With four units and two treated units, the ranks are always

1, 2, 3, 4.

- Under random assignment, the six possible treated rank sums are

$$\begin{array}{lll} 1 + 2 = 3, & 1 + 3 = 4, & 1 + 4 = 5, \\ 2 + 3 = 5, & 2 + 4 = 6, & 3 + 4 = 7. \end{array}$$

- Reference distribution: $\{3, 4, 5, 5, 6, 7\}$
- Therefore, boundary values give the largest p-value

$$\text{Boundary: } S = 5 \Rightarrow p = \frac{4}{6},$$

$$\text{Another possibility: } S = 7 \Rightarrow p = \frac{1}{6}.$$

Added: Quantile Effect (1): The null hypothesis

- Suppose that we have 100 students and want to test

$$H_0 : Q_{\{Y(1)-Y(0)\}}(0.95) \leq 10.$$

- **Interpretation:** This means that at least 95 of the 100 students have treatment effects of at most 10 points.
- Up to five students may have effects greater than 10.
- The null does not tell us **which students** these are, or what anyone's **exact treatment effect** is.

Added: Quantile Effect (2): Possible effect vectors

- Let $\tau = (\tau_1, \dots, \tau_{100})$ be a candidate vector of individual treatment effects.
- If τ is the true effect vector, we can recover each student's untreated potential outcome:

$$Y_i(0) = Y_i - T_i\tau_i.$$

- The tricky part: **many effect vectors satisfy the null**. For example,
 - Everyone's effect is exactly 10.
 - 95 students have effects of 10, and five have effects of 200.
- Both satisfy H_0 , but they can give different adjusted outcomes and test statistics.

Added: Quantile Effect (3): The hardest case to reject

- Recall the effect-increasing null case (pp.14-16)
 - The boundary gives the largest p-value. Rejecting it implies rejecting every other possibility allowed by that null.
- Use the same logic for the quantile null: among all effect vectors satisfying H_0 , find the one with the largest p-value.
- For our upper-tail rank-sum test, a common reference distribution means this is equivalent to finding the smallest test statistic:

$$S(\tau) = \sum_{i=1}^{100} T_i R_i(\mathbf{Y} - \mathbf{T} \circ \tau),$$

where $\circ$ denotes elementwise multiplication.

- If even this hardest case is rejected, we reject the quantile null.

Test Inversion: Procedure

- **Procedure**

- STEP 1: Define the range of treatment effect values
- STEP 2: For each value in the range, calculate the test statistic and compute the p-value
- STEP 3: Collect the range of p-values that is larger than α

- **Note:** Since the reference distribution does not depend on τ , you can simplify the procedure above using `wilcox.test()` on the treatment effect adjusted outcome

Added: Point Estimation (Hodges and Lehmann estimator)

- Assume the **population shift model**:

$$\Pr(Y_i(0) \leq y) = \Pr(Y_i(1) - \tau_0 \leq y)$$

- This is weaker than constant treatment effect. It says the distribution of $Y_i(1)$ is the same as that of $Y_i(0)$ shifted by τ_0

- For each candidate τ , define

$$S_\tau = \sum_{i=1}^n T_i R_i (\mathbf{Y} - \tau \mathbf{T}).$$

- We know that under complete randomization,

$$\mathbb{E}(S_\tau \mid \mathcal{O}_n) = \mathbb{E}\left[\sum_{i=1}^n T_i R_i (\mathbf{Y} - \tau \mathbf{T})\right] = \frac{n_1(n+1)}{2}.$$

- You have one equation and one unknown τ , so you can solve for τ
- Since you cannot write this as a closed form solution, you try the range of τ and get the average of the τ that gives the closest value to $\frac{n_1(n+1)}{2}$
- This is called the **Hodges and Lehmann estimator**

Added: Point Estimation (Hodges and Lehmann estimator)

- Notice that S_τ is a monotonically non-increasing function of τ .
 - As τ increases, we subtract more from treated outcomes, so treated units move down in the ranking and S_τ decreases.
- What value of τ should we choose?
 - If τ is **too small**, we have not removed enough of the treatment effect:

$$S_\tau > \frac{n_1(n+1)}{2}.$$

- If τ is **too large**, we have removed too much:

$$S_\tau < \frac{n_1(n+1)}{2}.$$

- Therefore, we choose the value of τ where S_τ **crosses** its expected value:

$$\frac{n_1(n+1)}{2}.$$

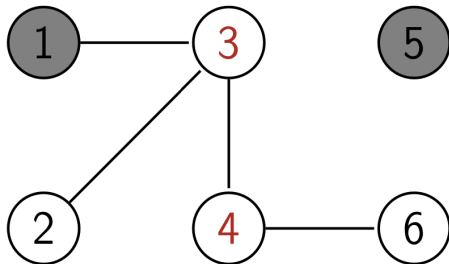
- Since S_τ changes in discrete steps, it may jump over this value. In that case, take the midpoint of the two values of τ around the crossing.

Beyond Rank Sum Test

- Procedure of permutation test does not depend on specific test statistic
 - STEP 1: Calculate test statistic with observed data
 - STEP 2: Permute the treatment assignment and calculate test statistic with permuted treatment assignment under the null
 - STEP 3: Compare the distribution of test statistic in STEP 2 (reference distribution) with the test statistic based on the observed data
- In Problem Set 2 Q2(b) and Q3, you might want to use other test statistic
- Read Chapter 5 of Imbens and Rubin (2015) for different test statistics

Conditional Randomization Test

- Let's understand the example from the class
 - Gray units: treated / White units: untreated
 - Focal units: unit 3 and 4



- Question: Are units 3 and 4 affected by the treatment of their friends?

Testing Spillover Effects

- Null Hypothesis: $Y_3, Y_4 \perp\!\!\!\perp T_1, T_2, T_5, T_6 \mid T_3, T_4$
 - Under the null, given the fixed value of T_3 and T_4 , the permutation of T_1, T_2, T_5, T_6 should not affect Y_3, Y_4
- Notice the difference / connection from permutation test
 - Permutation Test: Assume sharp null. Randomization is guaranteed by design
 - CRT: Randomization is justifiable under the null
- **Procedure**
 - STEP 1: Calculate test statistic $\text{Corr}(Y_{\text{focal}}, \bar{T}_{\text{friend}})$ on the observed data
 - STEP 2: Permute the treatment assignment T_1, T_2, T_5, T_6 given the original value of T_3, T_4
 - STEP 3: For each permutation, calculate the test statistic and create the reference distribution
 - STEP 4: Compare the observed test statistic with the reference distribution

Review Question: Setting

- Project STAR: does assignment to a **small class** affect high school graduation?
- Variables:
 - $T_i = 1$ for a small class; $T_i = 0$ for a regular class.
 - $Y_i = 1$ if student i graduates; $Y_i = 0$ otherwise.
 - n_1 treated students, n_0 controls, and $n = n_1 + n_0$.
- Assume **complete randomization**: every assignment with exactly n_1 treated students is equally likely.

Sharp null and test statistic

$$H_0 : Y_i(1) = Y_i(0) = Y_i \quad \text{for every } i, \quad S = \sum_{i=1}^n T_i Y_i.$$

Under H_0 , the total number of graduates $m = \sum_i Y_i$ is fixed; S counts graduates assigned to small classes.

Review Question 1: Mean of Fisher's Statistic

- Condition on the fixed potential outcomes

$$\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n.$$

- The statistic is

$$S = \sum_{i=1}^n T_i Y_i(1) = \sum_{i=1}^n T_i Y_i \quad \text{under } H_0.$$

Task

Derive the mean of Fisher's exact-test statistic under the sharp null:

$$\mathbb{E}(S \mid \mathcal{O}_n) = \frac{n_1 m}{n}.$$

Question 1 Tool: Expectation under Randomization

- **Linearity of expectation** does not require independent treatment indicators:

$$\mathbb{E}\left(\sum_i T_i Y_i(1) \mid \mathcal{O}_n\right) = \sum_i Y_i(1) \mathbb{E}(T_i \mid \mathcal{O}_n).$$

- Since T_i is binary,

$$\mathbb{E}(T_i \mid \mathcal{O}_n) = \Pr(T_i = 1 \mid \mathcal{O}_n).$$

Question 1 (Answer): Mean

$$\begin{aligned}\mathbb{E}(S \mid \mathcal{O}_n) &= \sum_{i=1}^n Y_i(1) \mathbb{E}(T_i \mid \mathcal{O}_n) \\ &= \frac{n_1}{n} \sum_{i=1}^n Y_i(1) \\ &= \frac{n_1}{n} \sum_{i=1}^n Y_i \quad (\text{Sharp null}) \\ &= \frac{n_1 m}{n}.\end{aligned}$$

- Please make sure that you understand every single step!

Review Question 2: Variance of Fisher's Statistic

Task

Derive the variance under the sharp null:

$$\mathbb{V}(S \mid \mathcal{O}_n) = \frac{mn_0n_1}{n(n-1)} \left(1 - \frac{m}{n}\right).$$

- 1 Show that, for $i \neq j$,

$$\mathbb{E}(T_i T_j \mid \mathcal{O}_n) = \frac{n_1(n_1 - 1)}{n(n - 1)}.$$

- 2 Expand the square using

$$\left(\sum_i X_i\right)^2 = \sum_i X_i^2 + \sum_{i \neq j} X_i X_j.$$

Question 2 Tool: Joint Treatment Probabilities

- If student j is treated, there are $n_1 - 1$ treatment slots among the remaining $n - 1$ students.

$$\begin{aligned}\mathbb{E}(T_i T_j \mid \mathcal{O}_n) &= \Pr(T_i = 1, T_j = 1 \mid \mathcal{O}_n) \\ &= \Pr(T_i = 1 \mid T_j = 1, \mathcal{O}_n) \Pr(T_j = 1 \mid \mathcal{O}_n) \\ &= \frac{n_1 - 1}{n - 1} \frac{n_1}{n}.\end{aligned}$$

Two identities for the next step

$$\mathbb{V}(S \mid \mathcal{O}_n) = \mathbb{E}(S^2 \mid \mathcal{O}_n) - \{\mathbb{E}(S \mid \mathcal{O}_n)\}^2.$$

Because treatment and graduation are binary, $T_i^2 = T_i$ and $Y_i(1)^2 = Y_i(1)$.

- The treatment indicators are dependent: treating one student slightly reduces another student's chance of treatment.

Question 2 (Answer): Second Moment

- Under H_0 , use $Y_i(1) = Y_i$ and expand:

$$S^2 = \left(\sum_{i=1}^n T_i Y_i \right)^2 = \sum_{i=1}^n T_i Y_i + \sum_{i \neq j} T_i T_j Y_i Y_j.$$

- Taking expectations gives

$$\mathbb{E}(S^2 \mid \mathcal{O}_n) = \frac{n_1 m}{n} + \frac{n_1(n_1 - 1)}{n(n-1)} \sum_{i \neq j} Y_i Y_j.$$

- There are m choices for the first graduate and $m - 1$ for the second, so $\sum_{i \neq j} Y_i Y_j = m(m - 1)$.
- As a result, we get

$$\mathbb{E}(S^2 \mid \mathcal{O}_n) = \frac{n_1 m}{n} + \frac{n_1(n_1 - 1)m(m - 1)}{n(n - 1)}.$$

Question 2 (Answer): Variance

Subtract the squared mean from Question 1:

$$\begin{aligned}\mathbb{V}(S \mid \mathcal{O}_n) &= \frac{n_1 m}{n} + \frac{n_1(n_1 - 1)m(m - 1)}{n(n - 1)} - \frac{n_1^2 m^2}{n^2} \\&= \frac{n_1 m \{n(n - 1) + n(n_1 - 1)(m - 1) - n_1 m(n - 1)\}}{n^2(n - 1)} \\&= \frac{n_1 m(n - n_1)(n - m)}{n^2(n - 1)} \\&= \frac{mn_0 n_1}{n(n - 1)} \left(1 - \frac{m}{n}\right).\end{aligned}$$

Review Question 3: Project STAR

	Small class	Regular class	Total
Graduate	754	892	1646
Not graduate	148	189	337
Total	902	1081	1983

Task

Replicate **both one-sided and two-sided p -values** in three ways:

- 1 Exact calculation using `fisher.test()`.
- 2 Normal approximation using the mean and variance above.
- 3 Monte Carlo approximation with 1,000 independent draws.

Question 3 Tool: From a Statistic to a p -Value

- **Exact reference distribution:** under H_0 ,

$$\Pr(S = k \mid \mathcal{O}_n) = \frac{\binom{m}{k} \binom{n-m}{n_1-k}}{\binom{n}{n_1}}.$$

The upper-tail p -value is $\Pr(S \geq s \mid \mathcal{O}_n)$, where $s = 754$ is the observed statistic.

- **Normal approximation:** standardize using Questions 1–2,

$$z = \frac{s - \mu}{\sigma}, \quad \mu = \frac{n_1 m}{n}, \quad \sigma^2 = \frac{m n_0 n_1}{n(n-1)} \left(1 - \frac{m}{n}\right).$$

Then $p_{\text{one}} \approx 1 - \Phi(z)$ and $p_{\text{two}} \approx 2\{1 - \Phi(|z|)\}$.

- **Monte Carlo:** keep outcomes fixed, repeatedly permute treatment, and estimate tail probabilities by their simulated frequencies.

Question 3 (Answer): Exact p -Values

```
star.tab <- matrix(c(754, 148, 892, 189), nrow = 2,  
  dimnames = list(  
    graduate = c("GRADUATE", "NOT GRADUATE"),  
    treat = c("SMALL CLASS", "REGULAR CLASS")))  
  
# One-sided: small classes increase graduation  
fisher.test(star.tab, alternative = "greater")$p.value  
# 0.2828931  
  
# Two-sided  
fisher.test(star.tab)$p.value  
# 0.5484734
```

Question 3 (Answer): Normal Approximation

```
n1 <- sum(star.tab[, "SMALL CLASS"])
n0 <- sum(star.tab[, "REGULAR CLASS"])
m <- sum(star.tab["GRADUATE", ])
n <- n0 + n1
s <- star.tab["GRADUATE", "SMALL CLASS"]

mu <- n1 * m / n
sigma2 <- m * n0 * n1 / (n * (n - 1)) * (1 - m/n)
z <- (s - mu) / sqrt(sigma2)

1 - pnorm(z)           # One-sided: 0.2627106
2 * (1 - pnorm(abs(z))) # Two-sided: 0.5254212
```

$$\mu \approx 748.710, \quad \sigma^2 \approx 69.397, \quad z \approx 0.635.$$

The observed count is about 0.64 standard deviations above the null mean. This calculation uses no continuity correction.

Question 3 (Answer): Monte Carlo Setup

- Reconstruct the student-level data from the table. Under the sharp null, $Y_i(0) = Y_i(1) = Y_i$.

```
treat <- c(rep(1, n1), rep(0, n0))
outcome <- c(rep(1, 754), rep(0, 148),
             rep(1, 892), rep(0, 189))

single_mc <- function(treat, outcome) {
  permuted <- sample(treat, replace = FALSE)
  sum(permuted * outcome)
}
```

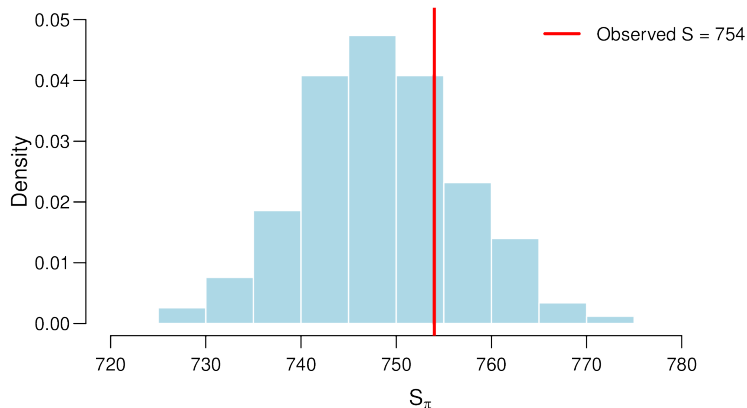
Question 3 (Answer): Monte Carlo p -Values

```
set.seed(02138)
nsim <- 1000
stats.perm <- replicate(nsim, single_mc(treat, outcome))

# One-sided: upper tail, including ties
p.upper <- mean(stats.perm >= s)
p.upper                                # 0.286

# Review handout's doubled-tail convention
p.lower <- mean(stats.perm <= s)
min(1, 2 * min(p.lower, p.upper)) # 0.572
```

Question 3 (Answer): Randomization Distribution



```
hist(stats.perm, col = "lightblue", freq = FALSE,  
     main = "", xlab = expression(S[pi]))  
abline(v = s, col = "red", lwd = 3)
```

The red line marks $s = 754$; 286 of the 1,000 draws have $S \geq 754$.

Question 3 (Answer): Compare the Results

Method	One-sided	Two-sided
Exact (<code>fisher.test</code>)	0.2829	0.5485
Normal approximation	0.2627	0.5254
Monte Carlo (1,000 draws)	0.2860	0.5720

- The normal approximation is close to the exact result.
- Monte Carlo results have simulation error; its two-sided rule also differs from `fisher.test()`'s rule.
- The observed graduation rates are $754/902 \approx 83.6\%$ and $892/1081 \approx 82.5\%$.

Conclusion

All three methods **fail to reject the sharp null** at $\alpha = 0.05$. This does not establish that every student's treatment effect is zero.

Appendix: Coding Exercise (Permutation Test)

- Let's try writing the code of Wilcoxon's rank sum test together!
- **Procedure**
 - STEP 1: Write a function to calculate test statistic (hint: use `rank` function)
 - STEP 2: Calculate test statistic with observed data
 - STEP 3: Permute the treatment and calculate test statistic under the permuted treatment assignment (use `sample` function)
 - STEP 4: Calculate p-value
- Note: `wilcox.test()` gives you **Mann-Whitney U test statistic**

$$U = \sum_{i=1}^n T_i R_i(Y_1, \dots, Y_n) - \underbrace{\frac{n_1(n_1 + 1)}{2}}_{\text{Mean}}$$

so without second term, the function gives the different value of test statistic

Appendix: Example Code (Permutation Test)

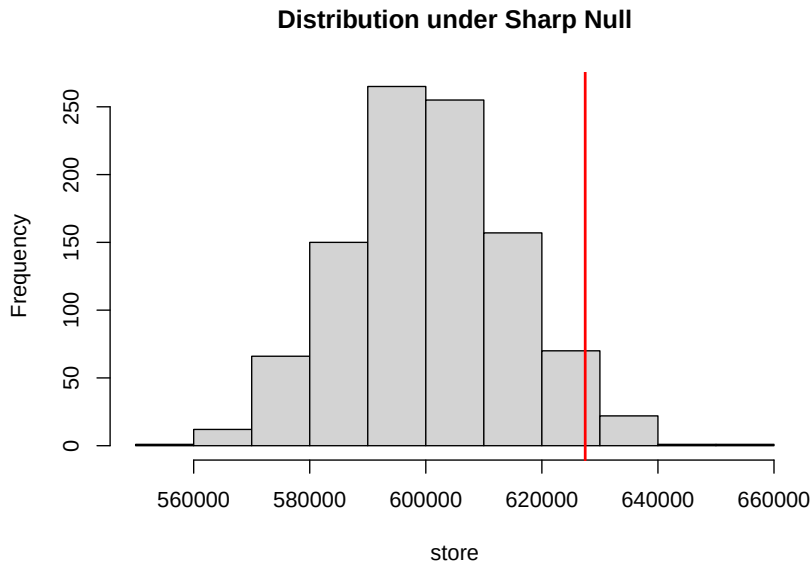
```
# function to calculate test statistics
calc_test_statistics <- function(treat, outcome) {
  n1 <- sum(treat)
  W <- sum(treat * rank(as.numeric(outcome)))
  U <- W - n1*(n1+1)/2
  return(U)
}
obs <- calc_test_statistics(treat = treat, outcome = Y)
```

Appendix: Example Code (Permutation Test)

```
nsim <- 1000 # nsim: Number of Monte Carlo Approximation
store <- rep(NA, nsim) #storage for each result

## Monte Carlo Approximation
for (b in 1:nsim){
  # permute treatment
  treat_sim <- sample(treat)
  # calculate test statistic
  store[b] <- calc_test_statistics(treat = treat_sim, outcome
}
```

Appendix: Reference Distribution



Appendix: With and without replacement

- You need to use **without replacement** (i.e., `replace = FALSE`)
 - With replacement: After you draw an item, you put it back into the population before drawing again.
 - Without replacement: After you draw an item, you do not put it back.
- This is because you need to fix the number of treated observation
 - If you put it back, then it is not guaranteed that the sum of treated people stays the same.
- Let's check how the code works.
 - With replacement, you might get more or less treated people.

```
# with replacement
set.seed(123)
sample(c(1,1,1,0,0,0), replace = TRUE) # 4 observations are treated
```

```
## [1] 1 0 1 1 1 0
```